Congruences and last digits

High school · number theory · Progressive exercises · 03

Exercise 1. Find the remainder of -37 modulo 8.

Exercise 2. Find the last digit of 7^{2026}.

Exercise 3. Prove that an integer square is never congruent to 2 modulo 3.

Exercise 4. Solve $3x \equiv 1 \pmod 7$.